

Problem 81. Let ABC be a triangle with a right angle at C and $AC < BC$. Let D be the foot of the perpendicular from C to AB . The circle centred at D with radius CD intersects BC at Q , and intersects the line AB at two distinct points E and F with F lying between A and B . Let QE intersect AC at P . Prove that $PE = QF$.

Problem 82. Find all pairs of positive integers (m, n) satisfying the following equation:

$$(m+1)^{m-1} + (m-1)^{m+1} = n^n.$$

Problem 83. A *knight* is a piece in the game of chess. It is known that a knight is allowed to move to a square on the chessboard that is two squares away horizontally and one square vertically, or two squares vertically and one square horizontally (i.e. the complete move therefore looks like the letter 'L').

An *open knight's tour* is a sequence of moves of a knight on a chessboard such that the knight visits every square exactly once.

Find all positive integers $n \geq 3$ for which there exists an open knight's tour on a $3 \times n$ chessboard.

(Note that the knight may start at any square on the chessboard and may not necessarily end at the square where it started.)

Problem 84. Let ABC be a triangle with a right angle at C , let I be its incentre, and let H be the foot of the perpendicular from C to AB . The incircle ω of the $\triangle ABC$ is tangent to the sides BC , CA , and AB at A_1 , B_1 , and C_1 , respectively. Let E and F be the reflections of C in the lines C_1A_1 and C_1B_1 , respectively, and let K and L be the reflections of H in the same lines. Prove that the circles A_1EI , B_1FI , and C_1KL have a common point.